Self-Similar Solutions of the Erdogan–Chatwin Equation

Nonlinear Diffusion, Similarity Theory, and Interdisciplinary Implications

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Abstract

The Erdogan–Chatwin equation is a nonlinear diffusion equation arising in buoyancy-enhanced transport phenomena. We study polynomial and self-similar solutions of the equation. It is shown that a quadratic polynomial ansatz yields only stationary affine solutions. We then derive self-similar solutions in both the weakly nonlinear and strongly nonlinear regimes. The equation is shown to interpolate between ordinary heat diffusion and cubic porous-medium dynamics. Connections are explored with physics, chemistry, biology, economics, finance, psychology, political science, warfare economics, artificial intelligence, and complex adaptive systems.

The paper ends with "The End"

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1 Introduction

Many natural and social systems exhibit diffusion-like behavior.

Examples include:

- heat conduction,
- molecular transport,
- groundwater movement,
- ecological migration,
- information propagation,
- capital flows,
- technological adoption,
- behavioral contagion,
- strategic influence diffusion.

The Erdogan–Chatwin equation represents a nonlinear generalization of the classical heat equation in which diffusivity depends upon the local gradient.

The governing equation is

$$\phi_t = \frac{\partial}{\partial x}\left[(1 + a\phi_x^2)\phi_x\right].$$

The parameter a controls the strength of nonlinear diffusion.

2 Physical Background

The equation was originally developed to model buoyancy-enhanced transport in horizontal fluid layers.

Density gradients generate:

1. buoyancy forces,
2. secondary circulation,
3. enhanced mixing,

4. nonlinear transport.

The diffusion coefficient becomes

$$D_{\text{eff}} = D(1 + a\phi_x^2).$$

Unlike ordinary diffusion, transport accelerates in regions of large concentration gradients.

3 Quadratic Polynomial Ansatz

Consider

$$\phi(x,t) = Ax^2 + Bxt + Ct^2 + Dx + Et + F.$$

Differentiation gives

$$\phi_x = 2Ax + Bt + D,$$

and

$$\phi_t = Bx + 2Ct + E.$$

Substituting into the Erdogan–Chatwin equation yields

$$Bx + 2Ct + E = 2A + 6aA(2Ax + Bt + D)^2.$$

Expanding the right-hand side gives

$$Bx + 2Ct + E = 2A + 24aA^3x^2 + 24aA^2Bxt + 6aAB^2t^2 + 24aA^2Dx + 12aABDt + 6aAD^2.$$

Matching coefficients implies

$$24aA^3 = 0.$$

Hence

$$A = 0.$$

The remaining coefficients satisfy

$$B = C = E = 0.$$

Theorem 1. *The only quadratic-polynomial solutions of the Erdogan–Chatwin equation are affine stationary solutions*

$$\phi(x,t) = Dx + F.$$

Proof. Coefficient comparison yields

$$A = B = C = E = 0.$$

Thus

$$\phi = Dx + F.$$

Since

$$\phi_t = 0$$

and

$$\frac{\partial}{\partial x}\left[(1 + aD^2)D\right] = 0,$$

the equation is satisfied identically.

□

4 Gradient Formulation

Define

$$u = \phi_x.$$

Differentiating the governing equation gives

$$u_t = \frac{\partial^2}{\partial x^2}(u + au^3).$$

Expanding yields

$$u_t = (1 + 3au^2)u_{xx} + 6au(u_x)^2.$$

This formulation is often more tractable than the original potential equation.

5 Similarity Analysis

Assume

$$u(x,t) = t^{-\alpha}F(\eta),$$

where

$$\eta = \frac{x}{t^\beta}.$$

Then

$$u_t = t^{-\alpha-1}\left(-\alpha F - \beta\eta F'\right),$$

and

$$u_{xx} = t^{-\alpha-2\beta}F''.$$

6 Weakly Nonlinear Regime

When

$$|au^2| \ll 1,$$

the equation reduces to

$$u_t = u_{xx}.$$

Balancing exponents gives

$$\beta = \frac{1}{2}.$$

Conservation of mass implies

$$\alpha = \beta.$$

Hence

$$\alpha = \beta = \frac{1}{2}.$$

The similarity solution becomes

$$u = t^{-1/2}F\left(\frac{x}{\sqrt{t}}\right).$$

The governing ordinary differential equation is

$$F'' + \frac{1}{2}\eta F' + \frac{1}{2}F = 0.$$

Its Gaussian solution is

$$F(\eta) = Ce^{-\eta^2/4}.$$

Therefore

$$u(x,t) = \frac{C}{\sqrt{t}}e^{-x^2/(4t)}.$$

Integrating gives

$$\phi(x,t) = C, \mathrm{erf}\left(\frac{x}{2\sqrt{t}}\right) + K.$$

7 Strongly Nonlinear Regime

For

$$au^2 \gg 1,$$

the equation becomes

$$u_t = 3a(u^2 u_x)_x.$$

Matching exponents yields

$$-\alpha - 1 = -3\alpha - 2\beta.$$

Thus

$$1 = 2\alpha + 2\beta.$$

Mass conservation implies

$$\alpha = \beta.$$

Consequently

$$\alpha = \beta = \frac{1}{4}.$$

The similarity form becomes

$$u = t^{-1/4}F\left(\frac{x}{t^{1/4}}\right).$$

Substitution yields

$$12a(F^2F')' + F + \eta F' = 0.$$

8 Similarity Exponents

Model	Similarity Exponent	Width Growth
Heat Equation	1/2	$t^{1/2}$
Erdogan–Chatwin (Weak)	1/2	$t^{1/2}$
Erdogan–Chatwin (Strong)	1/4	$t^{1/4}$
Porous Medium ($m = 3$)	1/4	$t^{1/4}$

Table 1: Similarity scaling laws.

9 A Phase Diagram

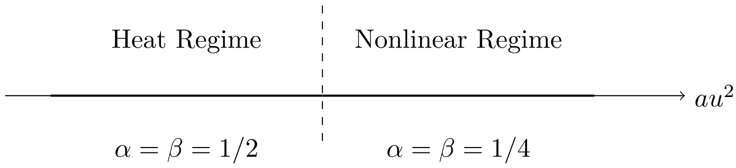

Figure 1: A Phase Diagram

10 Interpretations Across Disciplines

10.1 Physics

The equation describes nonlinear transport generated by buoyancy-driven enhancement of diffusion.

10.2 Chemistry

Chemical concentration gradients can generate nonlinear reaction-diffusion structures exhibiting Erdogan–Chatwin-type transport.

10.3 Biology

Population migration may accelerate in regions of high population pressure, creating nonlinear dispersal effects.

10.4 Economics

Let ϕ represent economic opportunity density.

Then

$$\phi_x$$

measures incentive gradients.

Migration and capital flows increase where opportunities differ strongly.

10.5 Finance

The field ϕ may represent liquidity.

Liquidity diffuses more rapidly where market dislocations become extreme.

10.6 Psychology

Ideas spread faster when cognitive dissonance creates large belief gradients.

10.7 Psychiatry

Behavioral contagion can exhibit nonlinear diffusion under sufficiently strong social pressures.

10.8 Political Science

Political influence spreads through nonlinear diffusion mechanisms.

Large ideological gradients often generate accelerated polarization.

10.9 Geopolitics

Influence, capital, technology, and migration diffuse through strategic networks.

The nonlinear term represents strategic amplification.

10.10 International Relations

Alliance structures can amplify diffusion of doctrines and norms.

10.11 Welfare Economics

Redistribution policies may be viewed as diffusion processes acting on utility gradients.

10.12 Warfare Economics

Military resources diffuse toward high-threat regions.

Threat gradients play the role of ϕ_x.

10.13 Military Operations

The equation resembles strategic redeployment dynamics under nonlinear threat amplification.

10.14 Computer Science

The equation provides a nonlinear message-passing model.

10.15 Artificial Intelligence

Gradient-dependent diffusion resembles adaptive information propagation in neural architectures.

10.16 Data Science

The model suggests nonlinear smoothing operators that preserve large-scale structure.

11 Algorithm

```
Input:
parameter a
initial profile u(x,0)

While t < T:

‘‘‘
Compute ux

Compute uxx

Update:

ut =
  (1 + 3 a u^2) uxx
  \+ 6 a u (ux)^2

Advance time
‘‘‘

Output:
u(x,t)
```

12 Statistical Interpretation

Suppose

$$u(x,t)$$

represents a probability density.
The similarity solutions determine the temporal evolution of variance.
For the heat regime,

$$\mathrm{Var}(t) \propto t.$$

For the nonlinear regime,

$$\mathrm{Var}(t) \propto t^{1/2}.$$

The latter exhibits substantially slower spreading.

13 Philosophical Interpretation

The equation embodies a general principle:

> The rate at which differences disappear depends on the magnitude of the differences themselves.

This principle appears throughout nature, society, and strategic systems.

14 Conclusion

The Erdogan–Chatwin equation interpolates between ordinary diffusion and cubic porous-medium dynamics.

Polynomial analysis shows that quadratic solutions collapse to affine stationary states. Similarity analysis reveals two universality classes:

1. Gaussian diffusion with exponent 1/2.
2. Nonlinear diffusion with exponent 1/4.

These results establish deep connections between nonlinear transport phenomena and a broad class of physical, biological, economic, strategic, and computational systems.

References

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Glossary

ϕ Potential field.

u Gradient field, $u = \phi_x$.

a Nonlinearity parameter.

η Similarity coordinate.

α Amplitude scaling exponent.

β Spatial scaling exponent.

Similarity Solution A solution invariant under scaling transformations.

Porous Medium Equation Nonlinear diffusion equation with density-dependent diffusivity.

Heat Equation Linear diffusion equation.

Effective Diffusivity Diffusion coefficient modified by nonlinear effects.

The End